

UP LT GRADE • COMPUTER SCIENCE MAINS

Digital Logic — Expected Numerical Questions with Full Solutions

One Worked Example From Every Question Type

1. Number System Conversion • 2. Complement Arithmetic • 3. Code Conversion
4. Boolean Algebra Simplification • 5. K-Map Simplification
6. Combinational Circuit Derivation • 7. MUX / Decoder Sizing
8. Memory Sizing • 9. Counter Design • 10. Flip-Flop Excitation • 11. Bit-Range Questions

Practice set • step-by-step solutions • exam-style presentation

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Q1. Convert Between Number Systems

Question

Convert the decimal number $(173)_{10}$ into (a) binary, (b) octal, and (c) hexadecimal.

(a) Decimal to Binary

$$173 \div 2 = 86 \text{ rem } 1$$

$$86 \div 2 = 43 \text{ rem } 0$$

$$43 \div 2 = 21 \text{ rem } 1$$

$$21 \div 2 = 10 \text{ rem } 1$$

$$10 \div 2 = 5 \text{ rem } 0$$

$$5 \div 2 = 2 \text{ rem } 1$$

$$2 \div 2 = 1 \text{ rem } 0$$

$$1 \div 2 = 0 \text{ rem } 1$$

Reading remainders bottom-to-top: **10101101**

(b) Binary to Octal

Group 10101101 into sets of 3 from the right: 010 101 101

010 = 2, 101 = 5, 101 = 5 $\rightarrow$ **$(255)_8$**

(c) Binary to Hexadecimal

Group 10101101 into sets of 4 from the right: 1010 1101

1010 = A, 1101 = D $\rightarrow$ **$(AD)_{16}$**

Final Answer

$$(173)_{10} = (10101101)_2 = (255)_8 = (AD)_{16}$$

Q2. Fractional Decimal to Binary Conversion

Question

Convert the decimal number $(0.375)_{10}$ into binary.

Method: Repeated Multiplication by 2

$0.375 \times 2 = 0.75 \rightarrow$ integer part **0**

$0.75 \times 2 = 1.50 \rightarrow$ integer part **1**

$0.50 \times 2 = 1.00 \rightarrow$ integer part **1**

Remainder becomes 0.00 — stop.

Reading integer parts top-to-bottom: **0.011**

Final Answer

$(0.375)_{10} = (0.011)_2$

Verify: $0 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3} = 0 + 0.25 + 0.125 = 0.375 \checkmark$

Q3. Subtraction Using 2's Complement

Question

Using 8-bit 2's complement arithmetic, compute $(45)_{10} - (78)_{10}$.

Step 1: Represent Both Numbers in 8-bit Binary

$45 = 00101101$

$78 = 01001110$

Step 2: Find 2's Complement of 78 (the number being subtracted)

1's complement of $01001110 = 10110001$

Add 1: $10110001 + 1 = 10110010$ (this is -78 in 2's complement)

Step 3: Add $45 + (-78)$

```
00101101 (45)
+ 10110010 (-78)
-----
11011111
```

Step 4: Interpret the Result

Leading bit is 1 → result is negative. Take 2's complement of 11011111 to find its magnitude:

1's complement: $00100000 \rightarrow$ add 1: $00100001 = 33$

Final Answer

$45 - 78 = -33$ (confirmed: result bit pattern 11011111 represents -33 in 2's complement)

Q4. BCD and Excess-3 Code Conversion

Question

Represent the decimal number $(47)_{10}$ in (a) BCD and (b) Excess-3 code.

(a) BCD (Binary Coded Decimal)

Convert each digit separately to 4-bit binary:

$4 \rightarrow 0100$ $7 \rightarrow 0111$

BCD of 47 = **0100 0111**

(b) Excess-3 Code

Excess-3 = BCD value + 3 for each digit:

$4 + 3 = 7 \rightarrow 0111$

$7 + 3 = 10 \rightarrow 1010$

Excess-3 of 47 = **0111 1010**

Final Answer

$(47)_{10}$ in BCD = 0100 0111 in Excess-3 = 0111 1010

Q5. Binary to Gray Code Conversion

Question

Convert the binary number 1011 into its equivalent Gray code.

Rule

MSB of Gray code = MSB of binary (copy as-is).
Each subsequent Gray bit = XOR of current binary bit and the previous binary bit.

Step-by-Step

Binary Bits	Operation	Gray Bit
1 (MSB)	copy directly	1
1, 0	$1 \oplus 0$	1
0, 1	$0 \oplus 1$	1
1, 1	$1 \oplus 1$	0

Final Answer

Binary 1011 → Gray Code **1110**

Q6. Simplify Using Boolean Laws

Question

Simplify: $F = A \cdot B \cdot C + A \cdot B \cdot \bar{C} + \bar{A} \cdot B \cdot C$

Step 1: Group the first two terms which share $A \cdot B$:

$$F = A \cdot B \cdot (C + \bar{C}) + \bar{A} \cdot B \cdot C$$

Step 2: Apply Complement Law ($C + \bar{C} = 1$):

$$F = A \cdot B \cdot (1) + \bar{A} \cdot B \cdot C = A \cdot B + \bar{A} \cdot B \cdot C$$

Step 3: Factor out B from both terms:

$$F = B \cdot (A + \bar{A} \cdot C)$$

Step 4: Apply the Absorption-type identity ($A + \bar{A} \cdot C = A + C$):

$$F = B \cdot (A + C)$$

Final Answer

$F = A \cdot B \cdot C + A \cdot B \cdot \bar{C} + \bar{A} \cdot B \cdot C$ simplifies to $F = B(A + C)$

Q7. 4-Variable K-Map Simplification

Question

Simplify $F(A,B,C,D) = \sum m(0, 1, 2, 5, 8, 9, 10)$ using a K-Map.

Step 1: Plot the K-Map

	CD=00	CD=01	CD=11	CD=10
AB=00	1 (m0)	1 (m1)	0 (m3)	1 (m2)
AB=01	0 (m4)	1 (m5)	0 (m7)	0 (m6)
AB=11	0 (m12)	0 (m13)	0 (m15)	0 (m14)
AB=10	1 (m8)	1 (m9)	0 (m11)	1 (m10)

Step 2: Form Groups

Group 1 (size 4): m0, m2, m8, m10 — all have B=0, D=0 → term = $\bar{B}\bar{D}$

Group 2 (size 2): m0, m1 — A=0, B=0, C=0 → term = $\bar{A}\bar{B}\bar{C}$

Group 3 (size 2): m1, m5 — B varies but A=0, C=0, D=1 → term = $\bar{A}\bar{C}D$

Group 4 (size 2): m8, m9 — A=1, B=0, C=0 → term = $A\bar{B}\bar{C}$

Optimizing: Groups 2 and 4 can be merged since both have $\bar{B}\bar{C}$ (A varies) → combined term = $\bar{B}\bar{C}$

This covers m0, m1, m8, m9 in one group of 4, which is better than two separate groups of 2.

Step 3: Final Groups Selected

Group A: m0, m2, m8, m10 → $\bar{B}\bar{D}$

Group B: m0, m1, m8, m9 → $\bar{B}\bar{C}$

Group C: m1, m5 → $\bar{A}\bar{C}D$

Final Answer

$$F = \bar{B}\bar{D} + \bar{B}\bar{C} + \bar{A}\bar{C}D$$

Q8. Derive Full Adder Expressions

Question

Construct the truth table for a Full Adder (inputs A, B, Cin) and derive simplified expressions for Sum and Carry-out.

Truth Table

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Deriving Sum (SOP from rows where Sum=1)

$$\text{Sum} = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in}$$

This expression cannot be reduced further by grouping — it directly equals the **3-input XOR**:

$$\text{Sum} = A \oplus B \oplus C_{in}$$

Deriving Carry-out (SOP from rows where Cout=1)

$$C_{out} = \bar{A}BC_{in} + A\bar{B}C_{in} + AB\bar{C}_{in} + ABC_{in}$$

Group $(\bar{A}BC_{in} + A\bar{B}C_{in}) = ABC_{in}(\bar{B}+B)$... more directly, group pairs sharing two variables:

$$= AB + BC_{in} + AC_{in} \quad (\text{each pair of inputs being 1 is enough to generate a carry})$$

Final Answer

$$\text{Sum} = A \oplus B \oplus C_{in} \quad C_{out} = AB + BC_{in} + AC_{in}$$

Q9. Multiplexer Sizing & Output Evaluation

Question

- (a) How many select lines are needed to build a 32-to-1 multiplexer?
(b) A 4-to-1 MUX has inputs $I_0=0$, $I_1=1$, $I_2=0$, $I_3=1$. If select lines $S_1S_0 = 11$, what is the output?

(a) Select Lines for 32-to-1 MUX

Number of inputs = 2^n , where n = number of select lines.

$32 = 2^5 \rightarrow n = 5$ select lines

(b) Evaluating 4-to-1 MUX Output

$S_1S_0 = 11$ selects input I_3 (since $S_1S_0=00 \rightarrow I_0$, $01 \rightarrow I_1$, $10 \rightarrow I_2$, $11 \rightarrow I_3$).

$I_3 = 1$, so output $Y = 1$

Final Answer

(a) 5 select lines needed (b) Output $Y = 1$ (I_3 selected)

Q10. Memory Address Lines & Chip Count

Question

- (a) How many address lines are needed for a 1024×8 memory chip?
- (b) How many such chips are needed to build an $8K \times 8$ memory system?

(a) Address Lines

$1024 = 2^{10}$ locations $\rightarrow$ **10 address lines needed** (data width of 8 needs 8 separate data lines, not address lines).

(b) Number of Chips for $8K \times 8$

$8K = 8192 = 2^{13}$ locations.

Each chip provides $1024 = 2^{10}$ locations.

Chips needed = $8192 / 1024 = \mathbf{8 \text{ chips}}$ (arranged side-by-side to extend the address range; data width already matches at 8 bits, so no chips are needed just for width in this case).

Final Answer

- (a) 10 address lines (b) 8 chips of 1024×8 needed to build $8K \times 8$ memory

Q11. Mod-6 Counter Design

Question

How many flip-flops are required to design a Mod-6 counter, and what is its state sequence?

Step 1: Find Minimum Flip-Flops

A counter with n flip-flops can count 2^n states.

We need at least 6 states $\rightarrow$ try $n=2$: $2^2=4$ (not enough). Try $n=3$: $2^3=8$ (enough).

Minimum flip-flops required = 3 (giving 8 possible states, of which only 6 are used).

Step 2: Write the State Sequence

000 $\rightarrow$ 001 $\rightarrow$ 010 $\rightarrow$ 011 $\rightarrow$ 100 $\rightarrow$ 101 $\rightarrow$ (back to) 000

States 110 and 111 are unused/invalid — the counter resets after reaching 101 (decimal 5), giving exactly 6 states (0 to 5).

Final Answer

3 flip-flops needed; sequence: 000, 001, 010, 011, 100, 101, then repeats.

Q12. JK Flip-Flop Excitation Table

Question

A JK flip-flop is currently at $Q=0$ and must transition to $Q(\text{next})=1$. What values of J and K are required?

JK Excitation Table (Reference)

Q (present)	Q(next)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Applying to This Case ($Q=0 \rightarrow Q(\text{next})=1$)

From the table, row $Q=0 \rightarrow Q(\text{next})=1$ requires: **J = 1, K = X (don't care)**

Reasoning: J=1,K=0 SETS the output to 1 (matches); J=1,K=1 TOGGLES $0 \rightarrow 1$ (also matches) — so K can be either value, hence "don't care."

Final Answer

J = 1, K = X (don't care)

Q13. Maximum Value Representable in N Bits

Question

Find the maximum positive decimal value representable using 8 bits in (a) unsigned form and (b) signed 2's complement form.

(a) Unsigned 8-bit

All 8 bits can be used for magnitude: max value = $2^8 - 1 = 256 - 1 = 255$
(Range: 0 to 255)

(b) Signed 2's Complement 8-bit

1 bit is reserved for the sign, leaving 7 bits for magnitude: max value = $2^7 - 1 = 128 - 1 = 127$
(Range: -128 to +127 — note the asymmetry, since 2's complement has only ONE representation of zero)

Final Answer

Unsigned max = 255 Signed (2's complement) max = 127, min = -128

— End of Practice Set —

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